

Space Force Data Fusion for Command and Control

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Sprint #1
9/10/2026 - 9/23/2026

Vision:

For the military personnel on the Vandenberg Space Force Base who need an efficient way to view geolocational data, the Data Fusion for Command and Control is a web-based application that is a data center that can help Vandenberg personnel complete tasks and get information more efficiently. Our product will ensure that military personnel is not wasting time or money by needing to login or access multiple places in order to get the information they need to do their job (routes, sites, personnel locations, etc).

Backlog:

- Setup [Next.js](#) monorepo
 - Develop a frontend for signup/login
- Set up CI and basic testing
- Research TAK

The screenshot displays a Jira Backlog interface. At the top, there's a header bar with a checkbox, a dropdown arrow, the text "Backlog (3 work items)", and three colored status boxes (grey, blue, green) each containing the number "0". To the right of these boxes is a "Create sprint" button. Below the header, there are three work items listed in a table-like structure. Each item has a checkbox on the left, followed by its ID and title. To the right of the title are three columns: a dropdown menu currently set to "BACKLOG", a minus sign icon, and a user icon. The third item, "HMSE - 21 Setup Next.js", is highlighted with a dark blue background. At the bottom left of the interface is a "+ Create" button.

☐	Backlog (3 work items)	0	0	0	Create sprint
☒	HMSE - 17 Research TAK	BACKLOG ▾	-	👤	
☒	HMSE - 9 Set up CI and basic testing	BACKLOG ▾	-	👤	
☒	HMSE - 21 Setup Next.js	BACKLOG ▾	-	👤	

+ Create

Sprint Goal

- Setup initial repo
 - Setup Jira, CPSE GitHub, and Discord server
- Research technologies
 - Research TAK
 - Research Aviation Weather
 - Research ARCGIS
 - Read Space Force Manual
- User stories
 - Explore needs of users
- Create DB schemas
 - Schemas for
 - Personnel
 - Location
 - Equipment
 - Custom Map Layer

HMSE Sprint 1 10 Sep – 24 Sep (11 work items)				0	0	0	Complete sprint	...
Complete the DB Schemas								
☒	HMSE-1	set up Jira project		Done	▼	-		
☒	HMSE-2	Figure out backend framework		Done	▼	-		
☐	HMSE-3	Write user stories		Done	▼	-		I
☒	HMSE-4	Setup Codebase (Page for aviation weather)	🔗	Done	▼	-		I
☐	☒ HMSE-5	Setup Cal Poly repo	🔗	Done	▼	-		...
☒	HMSE-6	Create group repo		Done	▼	-		
☐	HMSE-8	Create DB schemas	🔗	In Progress	▼	-	🔄	B
☐	HMSE-10	Figma prototype for landing page		Done	▼	-		A
☒	HMSE-11	Research ARCGIS Documentation		In Progress	▼	-		
☒	HMSE-12	Research https://aviationweather.gov/		In Progress	▼	-		H
☒	HMSE-19	Read Space Force manual		In Progress	▼	-		

Metrics:

Sprint Burn down (Jira)

[Chart](#)

Story points:

[Chart](#)

Demo(s):

Figma design: [here](#)

Aviation weather demo:

Retrospective:

What we did well

- Good communications
- Constantly occupied
- Peer review and feedback

What we could work on going forward

- Better time management
- Having more questions before the meeting
- Setting up more meetings
- Put in more time into research

Next Sprint:

- Setup [Next.js](#) monorepo
 - Develop a frontend for signup/login
- Set up CI and basic testing
- Research ARCGIS documentation
- Research TAK
- Read Space Force Manual
- Research Aviation Weather
- DB Schemas